BIT503 Fundamentals of Information Technology
Assessment 3

Weighting

30%

Learning outcomes

3 Apply the processes and procedures necessary for hardware and software diagnostic testing, maintenance, technical and customer support, and installation and configuration of systems and business applications.

4 Design and create a simple website, including interactive digital media, that demonstrates the basic principles of good interface design and usability.

Instructions

Complete and submit your assessment according to the Open Polytechnic's Assessments webpage. This includes information on academic integrity, word limits and referencing.

- Include your name, student number and the assessment number.
- Number your pages.

Submission

- Submit your assessment in two files:
  - 1 x Microsoft Word document
  - 1 x zipped website code files.
- Submit your work through your iQualify course.
- Emailed assessments will not be accepted.
- You will receive an automated notice following submission.

By submitting your assessment, you confirm that it is your own, original work.

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Instruction

Read tasks carefully and answer the questions. Your answers, where necessary, should be evidence-based and appropriately cited and referenced.

Task 1: Developing a website

Use HTML and CSS to develop your website. You will choose the topic of your website. It could be sport, culture, food or something else you are interested in. This website will cater to young adults aged 18–35.

Part 1: HTML and CSS

Create a new folder called **website-development**. The website must include the following features:

- Create a home web page called index.html, a registration web page called register.html and one additional web page of your choosing.
- Each page must contain a header, nav, content and footer.
    - The header must contain the name of your website.
    - The nav must contain links to your home, register and web pages.
    - The content section of your web page must contain relevant information to your chosen theme.
    - The footer must contain your first and last names and Assessment 3's due date. These must be centred.
- Include five appropriately sized, royalty-free and styled images with alt properties.
- Include an embedded YouTube video.
- Include a styled table containing relevant information to your chosen theme. Use CSS to style your table.
- Include either a styled ordered or unordered list of relevant information to your chosen theme.
- Include an external URL that opens in a new browser tab.
- Use colour consistently across all web pages.

In addition, you must:

- use one CSS file containing styles linked by all web pages
- separate your CSS file and image files into folders
- correctly indent your HTML and CSS code.

(20 marks)

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Part 2: HTML form validation and JavaScript

Create a new file called script.js. In the register.html web page, link the script.js file and create a form. The form must contain the following labels and inputs:

- first name
- last name
- email address
- mobile number
- password
- confirm password
- submit button.

When the submit button is clicked, check the following:

- HTML validation
    - All inputs (excluding the submit button) are not empty.
    - First and last names contain only letters.
    - The email address is in a valid format.
    - The mobile number contains 10 numbers.
    - The password has a minimum length of 6 and a maximum length of 12 characters.
- JavaScript
    - The confirmed password is the same as the password above. If they are not the same, display an appropriate message in a paragraph element. The text colour of this message must be red.
    - If all inputs are valid, display them in a paragraph element. The text colour of this message must be green.

Please include a ZIP file of your **website-development** folder in your assessment submission. Your website does not have to be hosted on a server.

(25 marks)

Part 3: Usability and accessibility

Discuss the usability and accessibility principles implemented in the design and development of your website. Specifically, refer to Nielsen's heuristics for usability and outline how your website addresses these principles. Additionally, detail how your website ensures accessibility for a diverse range of users. Cite and reference your sources appropriately.

(Word count guideline: 400 words)

(20 marks)

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Task 2: Customer and technical support

In Task 1, you developed a website but have decided you will not be providing ongoing support and maintenance and wish to hand over the website to someone else. To ensure a smooth handover, you need to provide detailed instructions for making specific changes. The instructions must cover how to:

- add another web page
- change the embedded YouTube video URL
- edit the styled table
- change the ordered or unordered list to an upper or lower Roman list
- change the footer element contents; that is, first and last names and Assessment 3's due date.

(25 marks)

Overall presentation and referencing

Marks will be awarded for overall presentation, including grammar, spelling, structure and flow, appropriate acknowledgements, citations and referencing (in APA style).

(10 marks)

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Marking schedule

Task 1: Part 1 (20 marks)	**8–10 marks**	**6–7.5 marks**	**5–6 marks**	**4–4.5 marks**	**1–3.5 marks**
Task 1: Developing a website Part 1: HTML and CSS	The website implements all features and is fully functional and without any errors.	The website implements almost all the features and is almost fully functional. There may be minor errors that do not impact the user experience.	The website implements most of the features and is mostly functional. There are minor errors that have some negative impact on the user experience.	The website implements some of the features. Several errors negatively impact the user experience.	The website barely implements the features. Many errors significantly impact the user experience.
	8–10 marks	**6–7.5 marks**	**5–6 marks**	**4–4.5 marks**	**1–3.5 marks**
	CSS and image files fully organised into folders, CSS correctly linked on all pages and excellent, consistent code indentation.	Well-organised CSS and image files, CSS properly linked in most pages and good indentation with minor issues.	CSS and image files are mostly separated, CSS linked in most pages, with adequate but inconsistent code indentation.	Some separation of CSS and image files, inconsistent CSS linking across pages and some code indentation.	Minimal separation of CSS and images into folders, inconsistent CSS linking and poor code indentation.
Task 1: Part 2 (25 marks)	**8–10 marks**	**6–7.5 marks**	**5–6 marks**	**4–4.5 marks**	**1–3.5 marks**
Task 1: Developing a website Part 2: HTML form validation and JavaScript	The register form is complete, accurate, and clearly laid out with suitable labels and inputs. It meets all usability and accessibility standards.	The register form is complete and clearly laid out with suitable labels and inputs. Minor inaccuracies may be present, but they do not impede legibility.	The register form is complete, but there are minor issues with the layout of the labels and inputs. These issues occasionally impede legibility.	The register form is mostly complete, but there are issues with the layout of the labels and inputs. These issues impede legibility, causing significant usability concerns.	The register form is minimally complete, with a poor layout of the labels and inputs. These frequently impede legibility and understanding, severely affecting usability and functionality.

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	12–15 marks	10–11.5 marks	7.5–9.5 marks	6–7 marks	1–5.5 marks
	The register form has comprehensive HTML validation and JavaScript.	The register mostly has HTML validation and JavaScript.	The register form has limited HTML validation and JavaScript.	The register form has limited HTML validation and/or JavaScript.	The register form has incomplete HTML validation and/or JavaScript.

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Task 1: Part 3	16–20 marks	13–15.5 marks	10–12.5 marks	8–9.5 mark	1–7.5 marks
Task 1: Developing a website Part 3: Usability and accessibility	Thoroughly discusses the usability and accessibility principles implemented in the website design and development. Clearly references Nielsen's heuristics for usability, detailing how the website addresses each principle. Provides comprehensive details on how the website ensures accessibility for a diverse range of users.	Clearly discusses the usability and accessibility principles implemented in the website design and development. References Nielsen's heuristics for usability, explaining how the website addresses each principle. Provides solid details on how the website ensures accessibility for a diverse range of users.	Adequately discusses the usability and accessibility principles implemented in the website design and development but lacks detail in some areas. References Nielsen's heuristics for usability but may not address all principles comprehensively. Provides adequate details on how the website ensures accessibility for a diverse range of users.	Briefly discusses the usability and accessibility principles implemented in the website design and development but provides insufficient detail. References some of Nielsen's heuristics for usability but lacks clarity or depth. Offers minimal details on how the website ensures accessibility for a diverse range of users.	Provides minimal discussion of the usability and accessibility principles implemented in the website design and development. Poorly references Nielsen's heuristics for usability. Offers minimal details on how the website ensures accessibility for a diverse range of users.
Task 2	**20–25 marks**	**16.5–19.5 marks**	**12.5–16 marks**	**10–12 marks**	**1–9.5 marks**
Task 2: Customer and technical support	Thoroughly provides detailed and clear instructions with screenshots for making specific changes to the website. Covers how to add another web page, change the embedded YouTube video URL, edit the styled table, change the list and change the footer element contents with precision.	Clearly provides instructions with screenshots for making specific changes to the website. Covers how to add another web page, change the embedded YouTube video URL, edit the styled table, change the list and change the footer element contents with clarity.	Adequately provides instructions with screenshots for making specific changes to the website but lacks detail in some areas. Covers how to add another web page, change the embedded YouTube video URL, edit the styled table, change the list and change the footer element contents but may miss some critical points.	Briefly mentions instructions for making specific changes to the website but provides insufficient detail. Covers how to add another web page, change the embedded YouTube video URL, edit the styled table, change the list and change the footer element contents with minimal clarity.	Provides minimal instructions for making specific changes to the website. Poorly covers how to add another web page, change the embedded YouTube video URL, edit the styled table, change the list and change the footer element contents.

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Presentation	8–10 marks	6.5–7.5 marks	5–6 marks	4–4.5 marks	1–3.5 marks
Marks will be awarded for overall presentation, including grammar, spelling, structure and flow, appropriate acknowledgements, citations and referencing (in APA style).	Presentation is flawless in all aspects: excellent grammar, logical structure, complete acknowledgements, and APA formatting for citations and references, including the reference list without error.	Presentation is with minor issues in grammar or structure, acknowledgements, and/or APA formatting for citations and references. Includes a reference list with only minor errors.	Presentation is satisfactory but has several noticeable issues in grammar, structure or APA formatting that moderately affect its quality. Includes a reference list with only minor errors.	Presentation has significant errors in grammar, structure and/or APA formatting. Includes a reference list with some errors.	Presentation is unsatisfactory, with major errors in all aspects, making it unreadable and lacking in academic rigour. Does not include a reference list.

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